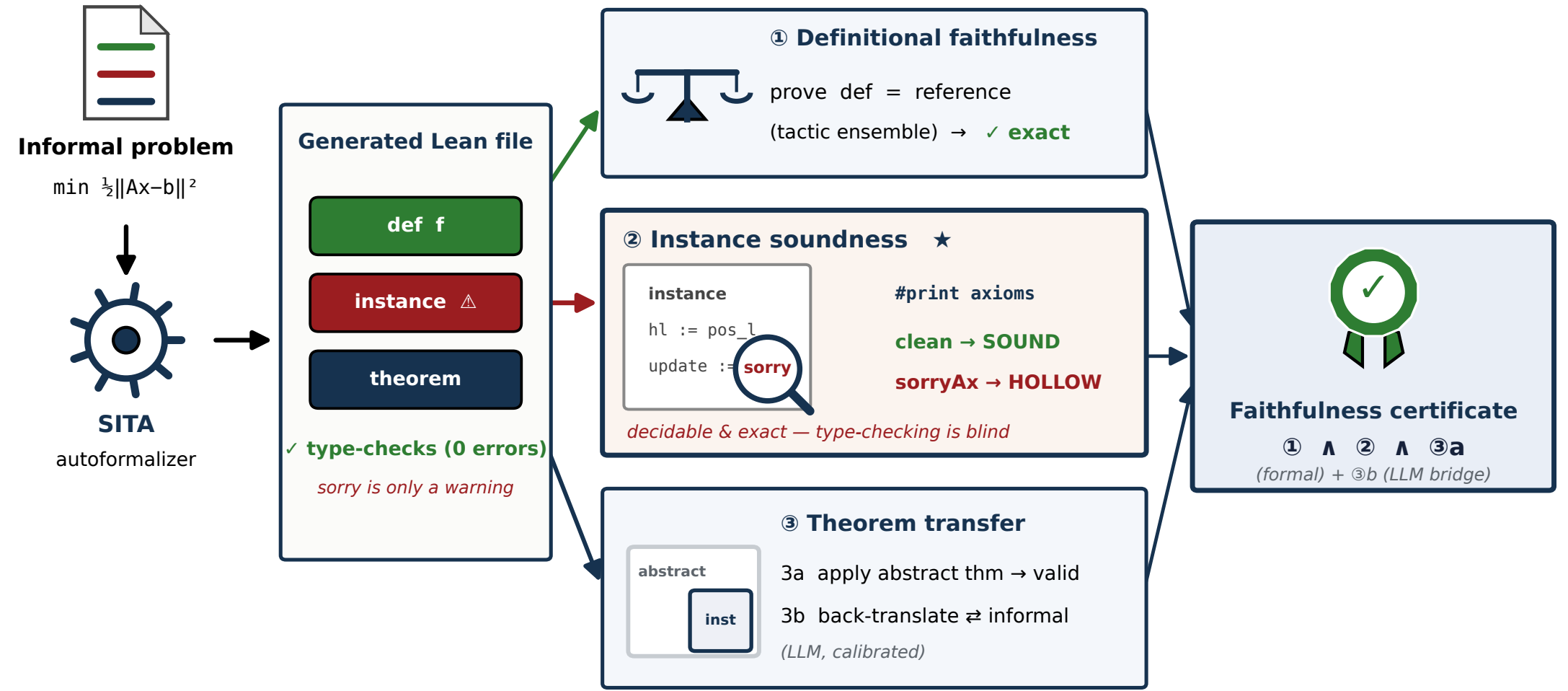
Informal problem
min ½‖Ax−b‖²
SITA
autoformalizer
Generated Lean file
def f
instance ⚠
theorem
✓ type-checks (0 errors)
sorry is only a warning
① Definitional faithfulness
prove def = reference
(tactic ensemble) → ✓ exact
② Instance soundness ★
instance
hl := pos_l
update := sorry
#print axioms
clean → SOUND
sorryAx → HOLLOW
decidable & exact — type-checking is blind
③ Theorem transfer
abstract
inst
3a apply abstract thm → valid
3b back-translate ⇄ informal
(LLM, calibrated)
Faithfulness certificate
① ∧ ② ∧ ③a
(formal) + ③b (LLM bridge)